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A college under Mapúa Malayan Colleges Laguna

Proposal for MotorPH Payroll and Inventory System

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MotorPH Requirements

Overview of the Need

MotorPH aims to develop an **end-to-end inventory** and **payroll system** to manage employee records and salary computation. However, due to project scope, complexity, and time constraints, the system will be developed **in phases**.

For **Phase 1**, the immediate business need is **not full payroll automation but** the creation of a **foundational payroll module** that can:

- Accurately present employee information
- Correctly compute weekly hours worked
- Automatically calculate gross and net weekly salaries
- Apply simplified government-mandated deductions

This phased development approach allows MotorPH to **validate payroll logic early**, reduce implementation risks, and establish a reliable base for future system expansion.

I. Identified Types of Requirements for MotorPH

A. Business Requirements

(Why is this project needed?)

MotorPH needs a basic payroll system to replace manual salary computation, reduce human error, and ensure consistent salary calculations for employees.

The project is required to:

- Improve efficiency in payroll processing
- Ensure compliance with simplified government deduction policies
- Provide a scalable foundation for future payroll and inventory features

B. Stakeholder Requirements

(Who needs the system and what do they expect?)

- **HR / Payroll Staff** need a system that can automatically compute salaries based on hours worked and deductions.
- **Management** expects accurate, consistent payroll outputs that can be reviewed weekly.
- **Employees** expect correct salary computation based on their actual work hours and fair application of deductions.

C. Solution Requirements – Functional

(What should the system do?)

The system shall:

- Display employee details in a prescribed format (employee number, name, birthday).
- Calculate the total number of hours worked by an employee per week using log-in and log-out data.
- Compute the gross weekly salary based on hours worked.
- Compute the net weekly salary by applying simplified SSS, PhilHealth, Pag-IBIG, and withholding tax deductions.
- Apply a 10-minute grace period before salary deductions are enforced.

D. Solution Requirements – Non-Functional

(How should the system perform?)

- **Usability:** The system should be easy to understand and use, even for users with limited technical skills.
- **Accuracy:** Salary computations must be mathematically correct and consistent with the provided deduction matrix.
- **Reliability:** The system should consistently produce correct results using the same input data.

E. Transition Requirements

(What needs to happen before or after the system goes live?)

- Payroll staff should be oriented on how to use the system before it replaces manual computation.
- Sample employee data should be tested prior to final submission to ensure correctness of outputs.

These requirements are temporary and only apply during deployment and testing.

F. Technical Requirements

(What technology does the system need?)

- The system must be capable of reading employee and attendance data from Google Sheets.
- The payroll computation logic should be implemented using programming code (as required in Phase 1).
- The system should be able to run on standard desktop or laptop environments.

G. User Interface (UI) Requirements

(What should the system look like to the user?)

- Employee information should be displayed in a clear, structured format.
- Salary computation results (hours worked, gross salary, deductions, and net salary) should be clearly labeled and easy to read.
- The interface should visually separate employee details from payroll results.

H. Hardware Requirements

(What physical devices are needed?)

- A standard desktop or laptop computer capable of running the payroll application.
- Internet access for retrieving employee data from Google Sheets (if required).

Summary

By identifying **business, stakeholder, solution, transition, technical, UI, and hardware requirements**, MotorPH's Phase 1 payroll system ensures that:

- The **right problem** is being solved
- **User and organizational needs** are clearly addressed
- The system is **technically feasible, usable, and scalable**

1. Phase 1 Objectives

- Display basic employee information
- Track employee attendance (log-in/log-out)
- Calculate weekly working hours
- Compute gross and net weekly salaries
- Applying simplified government-mandated deductions
- Ensuring accuracy and compliance with BIR and DOLE regulations

2. Phase 1 Scope

In Scope

- Employee details presentation (Employee Number, Name, Birthday)
- Timekeeping with 10-minute grace period
- Weekly hours worked calculation
- Gross salary computation
- Net salary computation with deductions (SSS, PhilHealth, Pag-IBIG, Withholding Tax)
- Code-based payroll logic implementation

Out of Scope

- Inventory management
- Sales transactions and analytics
- Mobile application development
- Mobile application
- Physical store system integration

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II. Effort estimate

The Effort Estimate breaks down the Phase 1 tasks of the MotorPH Payroll System into smaller, measurable actions and estimates the time required for each. This helps the team plan workloads realistically, identify potential bottlenecks, and allocate time effectively for tasks such as designing the interface, coding payroll calculations, and testing results. It serves as a foundation for creating a structured and achievable project timeline.

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Effort Estimation

GUIDE QUESTION	TASK	RESPONSE
What are the main tasks you need to do? (Insert additional rows if you have more than 10 tasks)	1	Visualization of requirements
	2	Plan the layout for employee information presentation
	3	Design a code structure for storing employee data
	4	Implement input for employee working hours
	5	Calculate total weekly hours worked
	6	Calculate gross weekly salary based on hours
	7	Apply basic deductions to compute net salary
	8	Display employee information with salary details
	9	Test and debug the code
	10	Document code and workflow
What do you need to know to accomplish these tasks? (Insert additional rows if you have more than 10 tasks)	1	Knowledge of required employee data (employee number, name, birthday)
	2	Basic UI/UX principles for table or console display
	3	Understanding of arrays, lists, or objects in Java
	4	Knowledge of reading user input and validating it
	5	Ability to sum multiple daily hour inputs
	6	Formula for hourly wage calculation
	7	Knowledge of deductions (tax, SSS, etc.)
	8	Combining data presentation and computed values
	9	Basic debugging techniques in Java
	10	Knowledge of documenting code and creating user guides
What additional help do you think you need to be able to do each task? (Insert additional rows if you have more than 10 tasks)	1	Access to sample employee data or dummy dataset
	2	Examples of simple UI design or presentation templates
	3	Examples of simple UI design or presentation templates
	4	Guidance on best input validation practices
	5	Sample formulas or pseudocode for calculating hours
	6	Clarification on overtime rules (if any)
	7	Examples of deduction formulas
	8	Sample formatted output for guidance
	9	Access to debugging tools or IDE features
	10	Template for documentation or reporting
Given the nature of the task, the knowledge you need, and any additional help you need, how long do you think it would it take you (in hours) to finish each task? (Insert additional rows if you have more than 10 tasks)	1	**5 hours** Review project requirments and scope
	2	**6 hours** Analyze payroll rules and government deduction
	3	**6 hours** Design payroll system workflow
	4	**5 hours** Create system visualization and wireframe
	5	**4 hours** Prepare employee data structure
	6	**6 hours** Implement attendance and timekeeping logic
	7	**5 hours** Calculate hours worked
	8	**5 hours** Compute gross salary
	9	**6 hours** Compute net salary with deductions
	10	**7 hours** Testing, debugging, and revisions
In this course, you are expected to dedicate 7.5 hours per week. Based on the total number of hours you estimated, how many weeks will you need to finish your outputs?	Based on the estimated total of 55 hours, I will need approximately 8 weeks to complete all required outputs while dedicating 7.5 hours per week, which aligns with the official project timeline in the Milestone 1 plan	

You may refer to this link to view the photo more clearly: [Effort Estimation & Project Plan]

III. Use Case Diagram

The Use Case Diagram illustrates the core functionalities of the **MotorPH Phase 1 Payroll System**. It shows how the **Payroll Officer** manages payroll processing by calculating hours worked, computing gross and net weekly salaries, and applying government-mandated deductions. The **Employee** interacts with the system mainly to view employee details. The diagram also highlights conditional behavior through late deductions, ensuring accurate and fair salary computation based on attendance records.

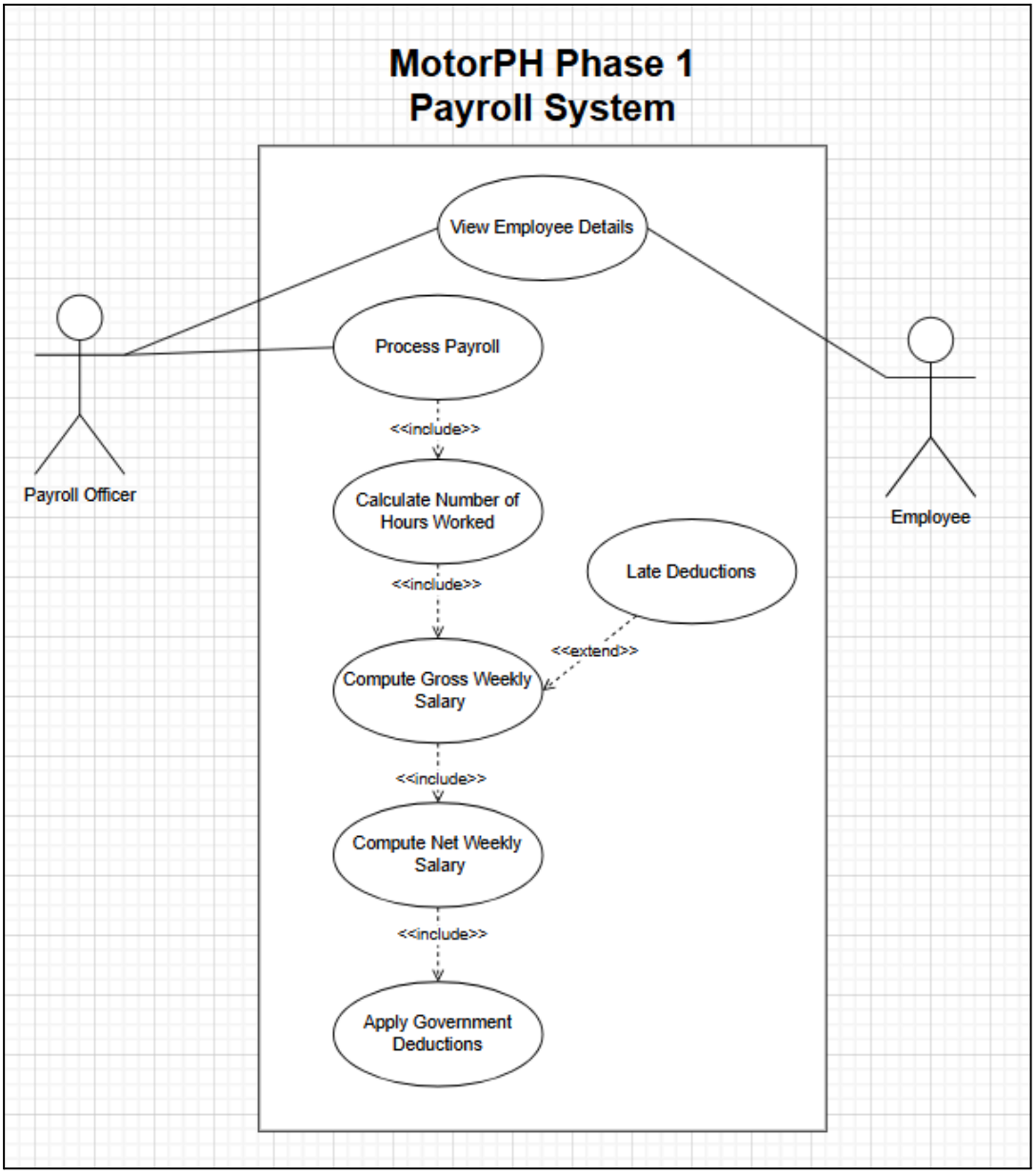
Actors

1. **Payroll Officer**
   – Responsible for processing payroll, calculating salaries, and applying deductions.
2. **Employee**
   – Can view their own employee details and payroll-related information.

Use Cases

1. View Employee Details
2. Process Payroll
3. Calculate Number of Hours Worked
4. Compute Gross Weekly Salary
5. Compute Net Weekly Salary
6. Apply Government Deductions
7. Late Deductions

Diagram



You may refer to this link to view the photo more clearly: [MotorPH Use Case Diagram]

IV. Wireframes

The following summary outlines the initial design layout for the MotorPH Payroll System, specifically designated for **Phase 1 MotorPH**.

The wireframe provides a visual blueprint of the MotorPH Payroll System's user interface, serving as a functional roadmap for employee and payroll management. It shows how menus, input fields, reports, and messages are arranged on screen, and the order in which users interact with the system. For instance, the interface transitions from a secure login gateway to comprehensive dashboards where administrators can view employee details, track precise attendance logs—including hours worked and late status—and manage complex payroll computations. Furthermore, the system includes a dedicated individual breakdown screen that provides employees with a transparent view of their gross salary, statutory deductions such as SSS, PhilHealth, and Pag-IBIG, and the final net weekly pay. These wireframes help ensure that the application is intuitive, user-friendly, and aligned with MotorPH's expectations before actual development begins.

Wireframes (Text Description)

1. Login Screen

- **Username field**: Placeholder text "enter username"
- **Password field**: Placeholder text "enter password"
- **Login Button**: Labeled "LOG IN"
- **Additional Features**: "Keep me logged in" checkbox and "Forgot password?" link

2. Employee Information Screen

- **Search and Sort**: Fields to search by employee and sort the list
- **Employee List Table**:
    - **Employee Number**
    - **Name**
    - **Birthday**
- **Action Button**: "Edit Employee Details"

3. Attendance (Weekly Hours Worked) Screen

- **Payroll Period**: Dropdown to select the specific date range (e.g., February 2-6, 2026)

- **Attendance Table**:
  - **Employee Number**
  - **Name**
  - **Date**: Specific date of log
  - **Log In / Log Out**: Timestamps for the work day
  - **Hours Worked**: Calculated daily duration (e.g., 7.92 Hours)

4. Payroll Computation Screen (Admin View)

- **Employee Payroll Table**:
  - **Employee List**: Number and Name
  - **Computed Weekly Hours**: Total hours worked for the period
  - **Weekly Gross Salary**: Base pay before deductions
  - **Deductions Breakdown**: Combined total of statutory and tax deductions
  - **Weekly Net Salary**: Final take-home pay
- **Status and Action**: Payment status (Distributed, Pending, Failed Delivery) and Edit/Delete buttons
- **Process Button**: "Process Payroll" button

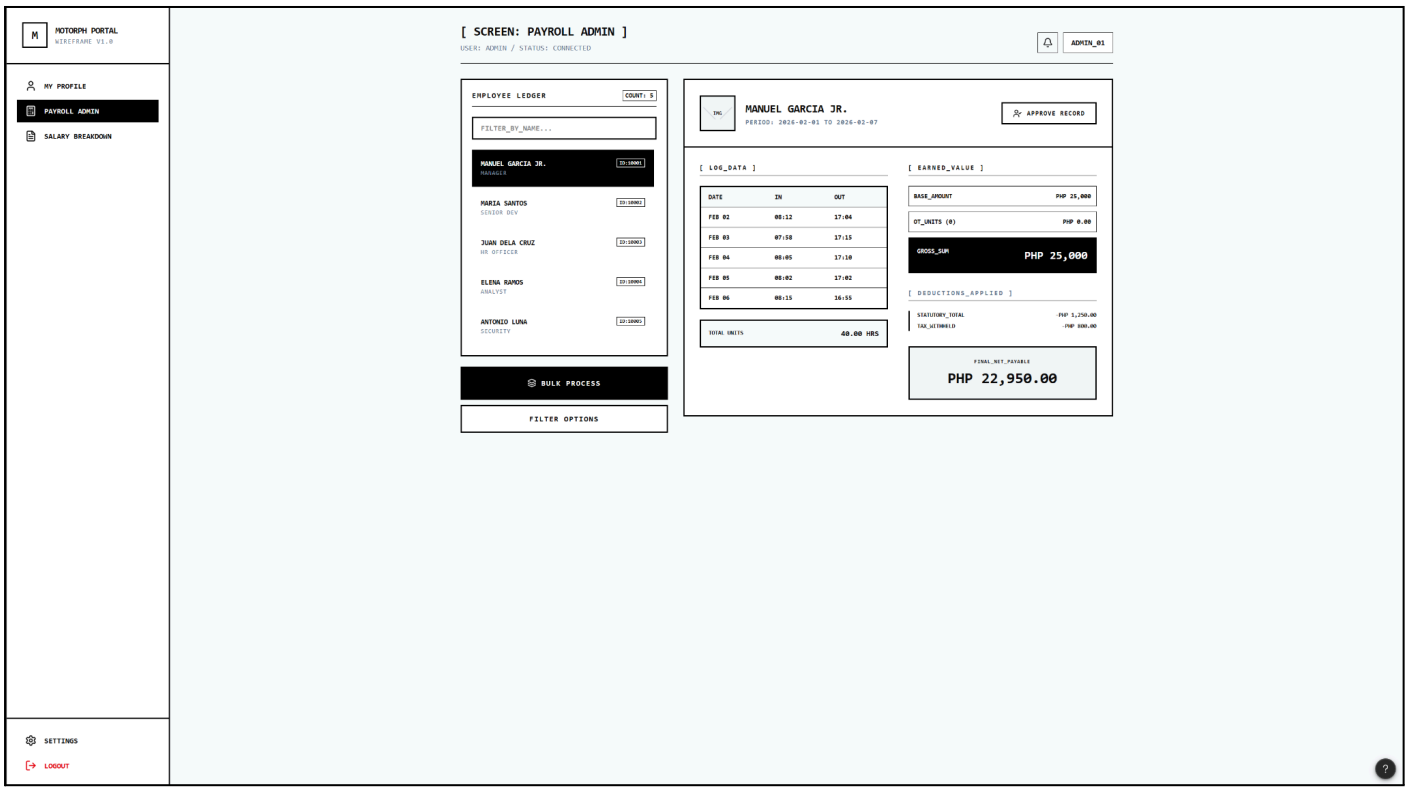
5. Salary Breakdown Screen (Individual View)

- **Employee Header**: Name and Position (e.g., Chief Executive Officer)
- **Time Records Card**: Lists daily Log In/Log Out times and the **Total Weekly Hours**
- **Calculated Earnings Card**:
  - **Basic Pay** and **Overtime**
  - **Gross Salary**
- **Deductions Card**:
  - **Statutory (SSS/PH/PI)**
  - **Withholding Tax**
  - **Other Deductions**
- **Total Net Pay**: Bold display of final weekly salary (e.g., ₱22,950.00)

Wireframes

Log In portal

HR Portal View



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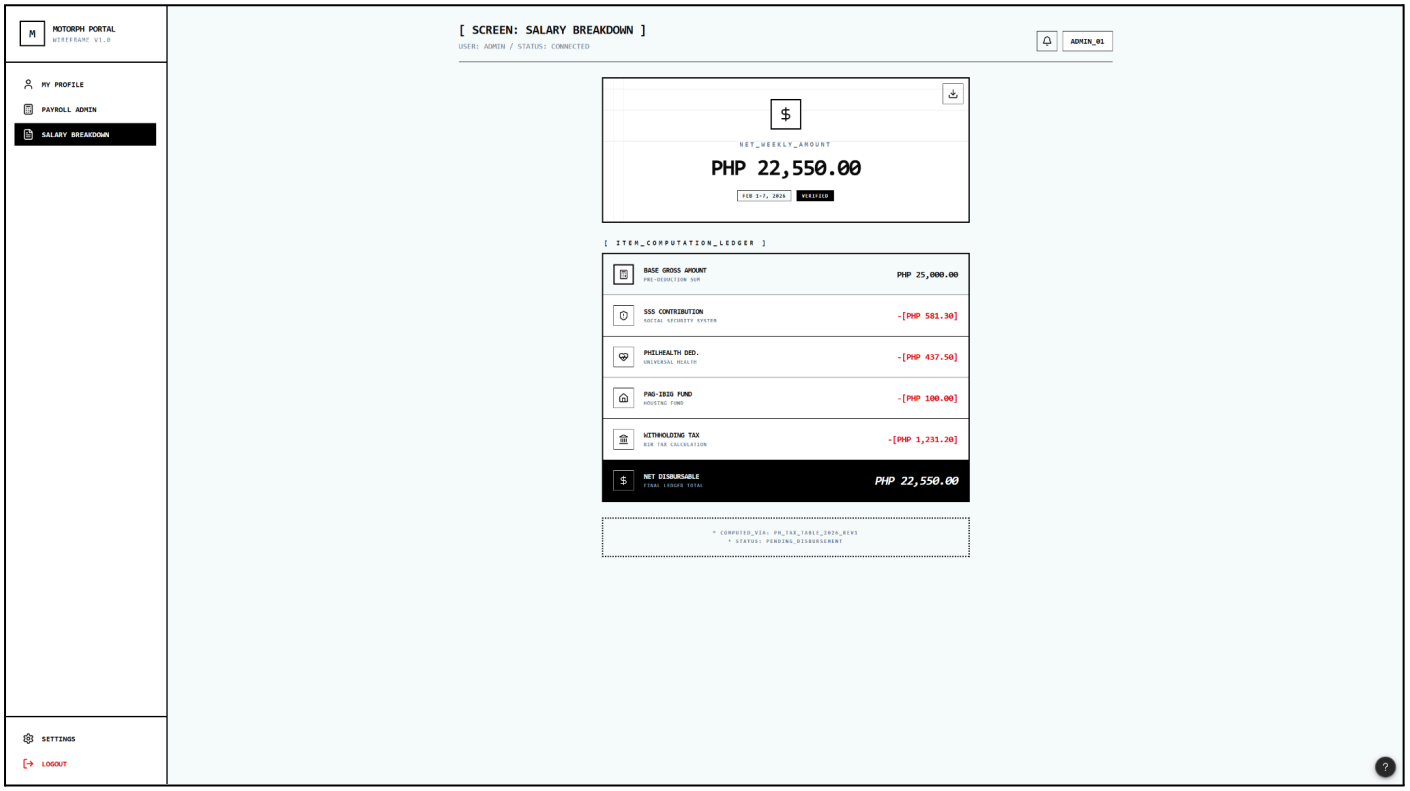

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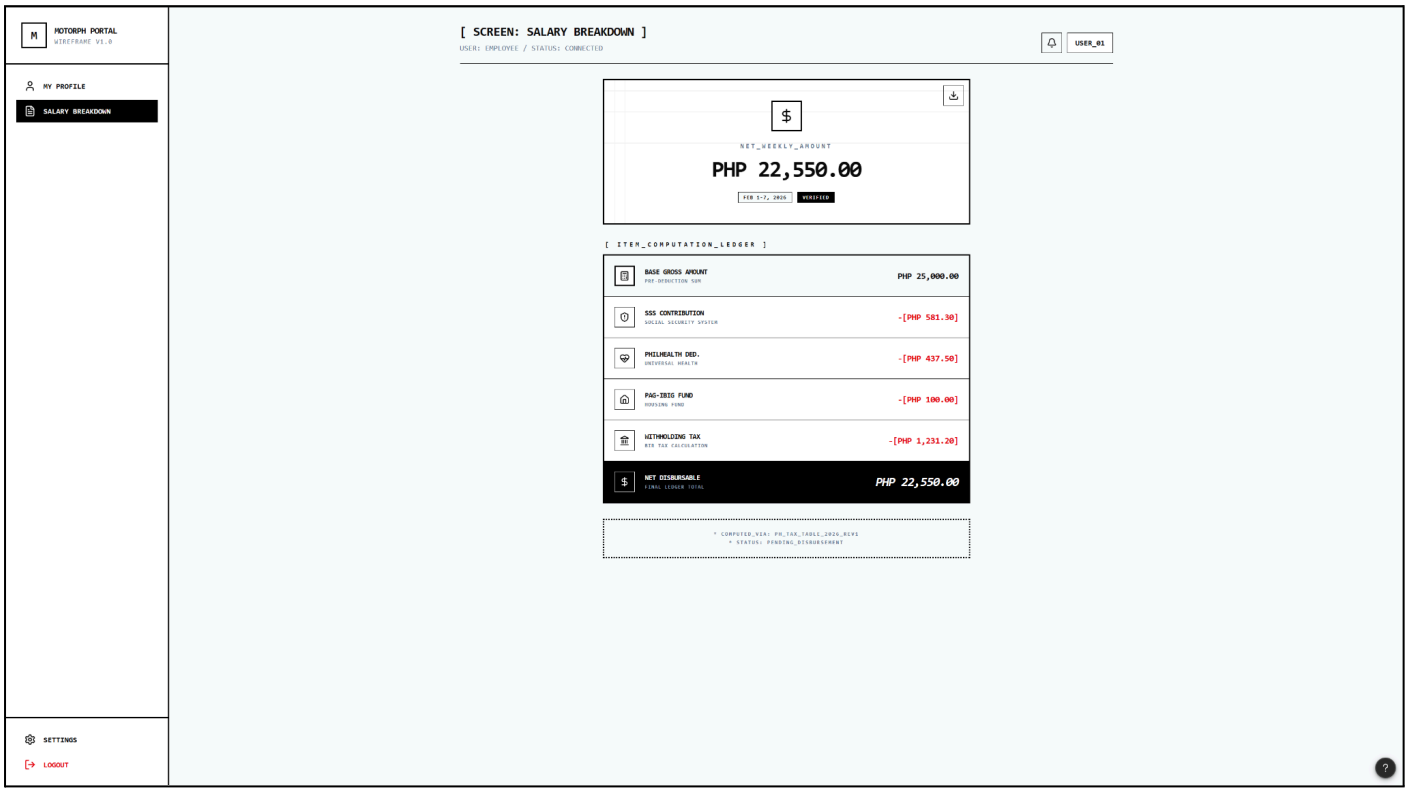
Employee Portal View



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V. Project Plan

The project plan organizes the MotorPH Payroll System development into structured tasks, timelines, and dependencies. It converts effort estimates into a roadmap, showing what needs to be done, in what order, and how long each task may take. This plan helps the team track progress, anticipate risks, and stay on schedule throughout the 11-week Phase 1 development period.

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Project Plan

TASK #	TASK	DEPENDENCY	OWNER	START DATE	END DATE	STATUS	ACTUAL START DATE	ACTUAL END DATE	NOTES
1	**Review and analyze project requirements**								
1.1	Review project requirements		Ephraim Elayda	Jan 5	Jan 7	Not Started			
1.2	Analyze payroll rules & deductions	1.1	Ashley Mackenzie Ramos	Jan 7	Jan 10	Not Started			
1.3	Define system scope & limitations	1.2	Ma. Irene Andrea Esguerra	Jan 8	Jan 13	Not Started			
2	**Design payroll workflow and logic**								
2.1	Draft payroll workflow	1.3	Princess Rey Ann Lene Romo	Jan 10	Jan 16	Not Started			
2.2	Research attendance rules	2.1	Ivone Christel Trinidad	Jan 15	Jan 19	Not Started			
3	**Create system wireframes**	2.1	Ephraim Elayda	Jan 18	Jan 22	Not Started			
3.1	Create system login wireframes	2.1							
3.2	Create payroll interface wireframe	3.1	Ashley Mackenzie Ramos	Jan 21	Jan 25	Not Started			
3.3	Design payment confirmation layout	3.2	Ma. Irene Andrea Esguerra	Jan 24	Jan 28	Not Started			
4	**Prepare employee and attendance data structure**	1.3							
4.1	Prepare employee data structure	1.3	Princess Rey Ann Lene Romo	Jan 27	Jan 31	Not Started			
4.2	Validate employee records	4.1	Ivone Christel Trinidad	Jan 30	Feb 3	Not Started			
5	**Develop attendance computation logic**	2.2							
5.1	Design attendance computation logic	2.2	Ephraim Elayda	Feb 2	Feb 6	Not Started			
5.2	Implement hours-worked computation	5.1	Ashley Mackenzie Ramos	Feb 5	Feb 9	Not Started			
6	**Develop attendance computation logic**	5.2							
6.1	Review salary computation formulas	5.2	Ma. Irene Andrea Esguerra	Feb 8	Feb 12	Not Started			
6.2	Design gross salary calculation	6.1	Princess Rey Ann Lene Romo	Feb 11	Feb 15	Not Started			
6.3	Design net salary calculation	6.2	Ivone Christel Trinidad	Feb 14	Feb 18	Not Started			
6.4	Integrate deduction logic	6.3	Ephraim Elayda	Feb 17	Feb 21	Not Started			
7	**Verify and validate payroll results**	6.4							
7.1	Verify payroll computation results	6.4	Ashley Mackenzie Ramos	Feb 20	Feb 24	Not Started			
7.2	Document payroll computation process	7.1	Ma. Irene Andrea Esguerra	Feb 23	Feb 27	Not Started			
8	**Testing and revision**	7.2							
8.1	Prepare test cases	7.2	Princess Rey Ann Lene Romo	Feb 26	Mar 2	Not Started			
8.2	Execute system testing	8.1	Ivone Christel Trinidad	Mar 1	Mar 5	Not Started			
8.3	Debug identified issues	8.2	Ephraim Elayda	Mar 4	Mar 8	Not Started			
8.4	Revise system based on feedback	8.3	Ashley Mackenzie Ramos	Mar 7	Mar 11	Not Started			
9	**Finalization and submission**	8.4							
9.1	Prepare final documentation	8.4	Ma. Irene Andrea Esguerra	Mar 10	Mar 14	Not Started			
9.2	Conduct final system review	9.1	Princess Rey Ann Lene Romo	Mar 13	Mar 17	Not Started			
9.3	Final submission & presentation preparation	9.2	Ivone Christel Trinidad	Mar 16	Mar 20	Not Started			

You may refer to this link to view the photo more clearly: [Effort Estimation & Project Plan]

Solutions Evaluation

Before finalizing the technical approach for MotorPH's Phase 1 Payroll System, multiple solution alternatives were evaluated to determine the most appropriate option based on scope, budget (PHP 2,500), timeline (11 weeks), and system requirements.

Option 1: Spreadsheet-Based Payroll Automation (Google Sheets / Excel)

Description

Payroll computation could be implemented directly within Google Sheets using formulas, conditional logic, and built-in functions to calculate hours worked, gross salary, and deductions.

Advantages

- Minimal development effort
- No additional software development required
- Easy to implement within a short timeframe
- Low cost

Limitations

- Limited scalability for future phases
- Reduced control over validation and automation logic
- Higher risk of accidental formula modification
- Not ideal for implementing structured login and system workflows

Evaluation

While this option is cost-effective and simple, it does not provide a structured system architecture suitable for long-term expansion into a full inventory and payroll system.

Option 2: Web-Based Payroll System (HTML/CSS/JavaScript + Database)

Description

Developing a web-based application accessible through a browser, connected to a database for employee and attendance records.

Advantages

- Accessible from multiple devices
- Scalable for future integration with inventory module
- Easier multi-user access
- More aligned with enterprise systems

Limitations

- Requires hosting and possible additional costs
- Higher development complexity
- Increased security considerations
- May exceed Phase 1 timeline constraints

Evaluation

Although scalable and future-ready, this option introduces additional infrastructure requirements and technical complexity that may not align with the limited scope and budget of Phase 1.

Option 3: Desktop-Based Payroll Application (Chosen Solution)

Description

A standalone desktop application developed using structured programming logic to handle employee data presentation, attendance computation, and payroll calculations.

Advantages

- Works offline
- Controlled development environment
- Lower infrastructure cost
- Easier to manage within 11-week timeline
- Sufficient for Phase 1 foundational payroll logic

Limitations

- Limited remote access
- Multi-user functionality may require enhancement in future phases

Justification for Selected Approach

The desktop-based payroll application was selected because it:

- Aligns with the PHP 2,500 budget constraint
- Fits within the 11-week development timeline
- Meets all Phase 1 functional requirements
- Provides structured system behavior beyond simple spreadsheet automation
- Establishes a stable foundation for future migration to a web-based or enterprise solution

This approach balances practicality, scalability, and technical feasibility while minimizing risk during the initial development phase.

Conclusion

The evaluation of alternative solutions demonstrates that the chosen architecture was not arbitrary but based on careful consideration of cost, complexity, scalability, and project constraints. By selecting a desktop-based solution for Phase 1, MotorPH ensures controlled development, manageable risk, and a scalable base for future system expansion.

Project Scope

The project focuses on developing a **basic payroll system (Phase 1)** that manages employee information, records working hours, and computes weekly salaries. The system covers employee data storage, attendance tracking, gross salary computation, basic deductions, and net salary output.

Tasks and Activities

- Review payroll requirements and rules
- Design the payroll workflow and system layout
- Prepare employee data structure
- Implement hours-worked computation
- Calculate gross and net weekly salary
- Apply basic deductions
- Test, debug, and finalize the system

Resources Needed

- **Tools:** Java (NetBeans), Google Sheets
- **People:** Developer, tester, HR reference
- **Data:** Sample employee and attendance records

Roles and Responsibilities

- **Project Manager:** Oversees progress and ensures milestones are met
- **Developers:** Design and implement payroll system features
- **Testers:** Perform testing and validation
- **HR Personnel:** Provide payroll rules and verify accuracy

Risk Management

Identified Risks

- Payroll calculation errors
- Development or testing delays
- Changes in deduction policies

Mitigation Strategies

- Regular testing and validation
- Built-in schedule buffer
- Use of simplified, fixed deduction rules

Budget Summary

The Phase 1 budget focuses on development, testing, and documentation using free tools.

Category	Purpose	Cost (PHP)
Software & Design Tools	NetBeans, JDK, Google Sheets, Figma	Free
Hardware	Existing laptop/PC	0
Internet	Research and collaboration	1,500
Contingency	Minor unforeseen expenses	1,000
Total Estimated Budget		**2,500**

Free and widely available tools are used to minimize costs while ensuring functional system development. A contingency fund supports unexpected needs.

Communication Plan

- **Daily:** Team coordination via group chat
- **Weekly:** Progress meetings and status reports
- **Milestones:** Design and workflow reviews
- **Final Week:** System presentation and evaluation

Escalation and Documentation

- Issues are discussed within the team first
- Unresolved concerns are raised during weekly meetings
- Critical issues are escalated to the instructor
- All documents, meeting notes, and revisions are stored in a shared Google Drive

Conclusion

This project plan outlines the successful development of **Phase 1 of MotorPH's Payroll System**, designed to support the company's growth as it expands from an online business to physical store operations across the Philippines. By focusing on accurate employee information management, attendance tracking, salary computation, and government-mandated deductions, the system establishes a strong foundation for efficient payroll processing.

The phased approach ensures that core payroll functionalities are delivered within the **11-week timeline while** allowing flexibility for future enhancements. Upon completion of Phase 1, MotorPH will have a reliable and compliant payroll system that improves operational efficiency, minimizes manual errors, and supports informed decision-making. This system will also serve as the groundwork for future phases, including inventory management and advanced reporting, as MotorPH continues to scale its operations.